Gravity-Based Hydropower Storage System

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Abstract— With technological advancements, solar energy systems have become commonplace in our lives. However, storing and utilizing the electricity generated by these systems at night poses a challenge. Therefore, we developed a liquid energy storage device to store and utilize this converted energy. During the day, the solar panels within the liquid energy storage device receive solar energy and convert it into electricity, driving a water pump to pump water from a lower location to a higher location. At night, the liquid energy storage device automatically releases the pumped water, converting potential energy into kinetic energy, which then flushes the pump and generates electricity.

Keywords— Gravity-based energy storage, solar energy, potential energy conversion, servo motor control, remote sensing

1. Background

Energy storage has always been a core issue driving innovation in the energy sector. With ever-increasing energy demand and increasingly severe environmental challenges, finding efficient and reliable energy storage solutions is becoming increasingly urgent. Liquid energy storage, as an emerging technology, offers new hope for the energy storage field. Liquid energy storage plays a crucial role in ensuring the stability and sustainability of energy supply. Solar panels are installed in many areas, but daytime energy storage remains a bottleneck. We believe that liquid energy storage has the potential to play a significant role in the energy system, building a more solid foundation for a sustainable energy future. To this end, we have designed a system that uses water potential to store energy, preserving the energy received by solar panels.

Figure 1

Furthermore, our device is primarily intended for use in the Yunnan-Guizhou Plateau, where sufficient terrain exists to implement the system. Furthermore, this device can help address the energy storage challenges faced by the plateau.

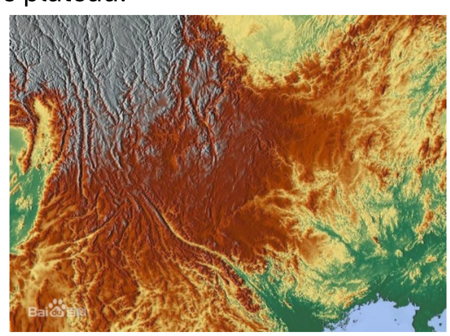

Figure 2

2. Research Objectives

Our research focuses on two key areas.

1) We will demonstrate the feasibility of liquid energy storage by developing a simple liquid energy storage system and combining it with remote and local monitoring methods.

2) We will comprehensively monitor the internal operating conditions of the liquid energy storage system to demonstrate its efficiency.

These steps aim to provide practical evidence for the practical application of liquid energy storage technology.

3. System Structure

3.1 The R&D process Overview

- Design the overall structure
- Select components, including controllers, sensors, solar panels, actuator systems, input systems, etc.
- Design from scratch
- Draw a complete design, design the installation, and install the circuitry
- Write the software program

3.2 Sensor Selection

Sensors include brightness sensors, water level sensors, tilt sensors, and more to ensure the proper operation of the entire system.

3.3 Main Control Board Selection

Our controller utilizes an open, highly integrated design to facilitate connection to sensors and control devices. It offers the following features:

- Adopts an advanced CGF control module for low power consumption and excellent compatibility
- Operating voltage: 7.5–9 VDC
- Features two high-torque motor drive ports and two servo motor drive ports
- Eight sensor input ports and multiple output ports for LEDs, LCDs, and more
- Equipped with a mini USB port

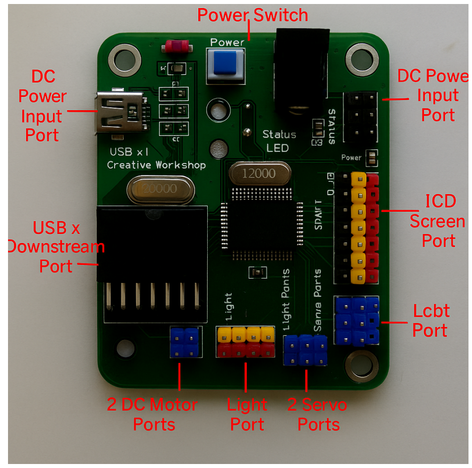

Figure 3

4. Production Process

The entire installation consists of a support system, two upper and lower water containers, a solar system, a simulated solar energy system, a power generation system, and a detection system.

The support system uses a metal frame with wooden planks in the middle to ensure structural stability. Two containers are mounted on the bottom and top planks, respectively. The solar panels are relatively long, allowing for easy extension to the required locations. The power generation system uses water flow to impact an impeller, driving a motor that generates electricity and illuminates an LED. In the detection system, a water level sensor detects the water level in the upper container. If it exceeds a certain level, an alarm is automatically triggered and a tilt switch is activated to disconnect the solar circuit, stopping the water pump. A brightness sensor detects nighttime and activates a metal high-torque servo motor to a certain angle, turning the faucet on. During daytime, the faucet is closed and filled with water.
The majority of the materials are wood, with some plastic used. Screws, nuts, and hot melt

adhesive are used for fastening. Some materials are made of wood. Acrylic is machined into the desired shape.

4.1 System Structure

The hot glue guns, screws and nuts and other tools are mainly used to build our creations.

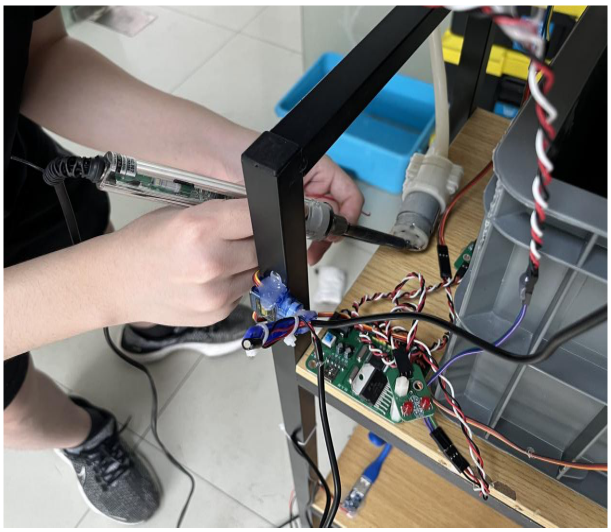

Figure 4

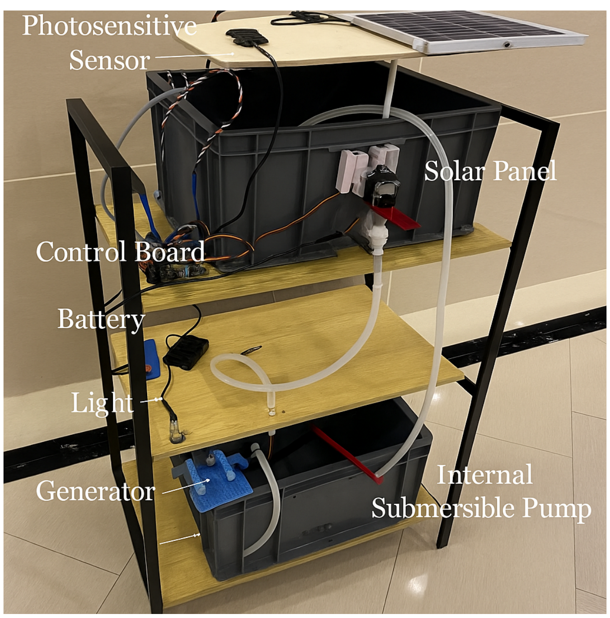

Figure 5

4.2 Remote Control

To realize automated environmental control, we developed a sensor-driven logic program using a graphical programming environment. The system is divided into two primary modules: Water Level Detection and Light Intensity Detection, each controlling corresponding servo motors based on sensor input values.

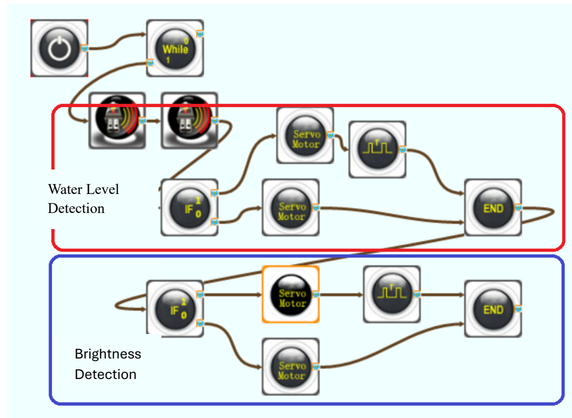

Figure 6

4.2.1 Water Level Detection Module

The top section of the program (highlighted in red in the diagram) is responsible for water level monitoring and control. This process begins within a While loop that ensures continuous real-time execution. Two ultrasonic sensors are used to detect the height of water. Once the signal is received, a conditional block (IF_0) evaluates whether the water level is below the predefined threshold. If true, the logic activates the servo motor to open the water valve and initiate water inflow. A control block then ensures the motor runs for a specific pulse duration before terminating the action with an END command. If the water level is deemed sufficient, the system skips this routine and awaits the next loop iteration.

4.2.1 **Light Intensity Detection Module**

The lower part of the program (highlighted in blue) is dedicated to light sensing. A photoresistor (light sensor) continuously detects ambient brightness. A similar conditional judgment (IF_0) compares the light intensity against the desired value. If the light is deemed too dim (e.g., during evening or low sunlight), the servo motor activates the lighting system or opens a light-blocking panel. The logic again uses a timed pulse block followed by an END command to manage operation duration.

4.2.3. System Behavior

This dual-sensing setup ensures that the smart agricultural or environmental setup remains responsive to both hydration and lighting conditions. The While loop at the start ensures perpetual monitoring, while the separate execution branches keep motor activation independent and efficient.

5. Product Description

Our liquid energy storage device consists of a water level sensor, a photosensor, a solar panel, a submersible pump, a generator, a servo motor, and a light. Turn on the power, press the reset and run buttons one by one, and place an incandescent lamp on the solar panel to simulate the sun to generate electricity. The generated electricity pumps water from a lower location to a higher location.

After a certain amount of water has been pumped, cover the photosensor with an object to simulate nighttime. The device automatically releases the water from the higher location, converting potential energy into kinetic energy, which then flushes the generator impeller. The large gear at the rear drives the small gear of the generator motor, accelerating the generation of electricity.

6. Experimental Results and Analysis

Throughout the entire process, even though the mechanical structure was consuming considerable time and effort, the program executed flawlessly on the hardware and achieved the desired results. Initially, a solenoid valve was used to release the water flow, but this was very power-intensive. After repeated trials, we adopted a servo motor to control the faucet opening and closing.

7. Conclusion

To address the current limitations of solar energy, which can only be used during the day, and the abundant water resources in southern China, our liquid energy storage device uses water from lower locations to store solar energy, which then flows to higher locations for use at night. This improves the utilization rate of both solar and water resources.

In the future, we can increase the height of the device, increasing its gravitational potential energy and thus increasing its power output. We also hope to miniaturize the device, allowing it to be used not only in large reservoirs and factories but also in homes, maximizing its potential.

8. References

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